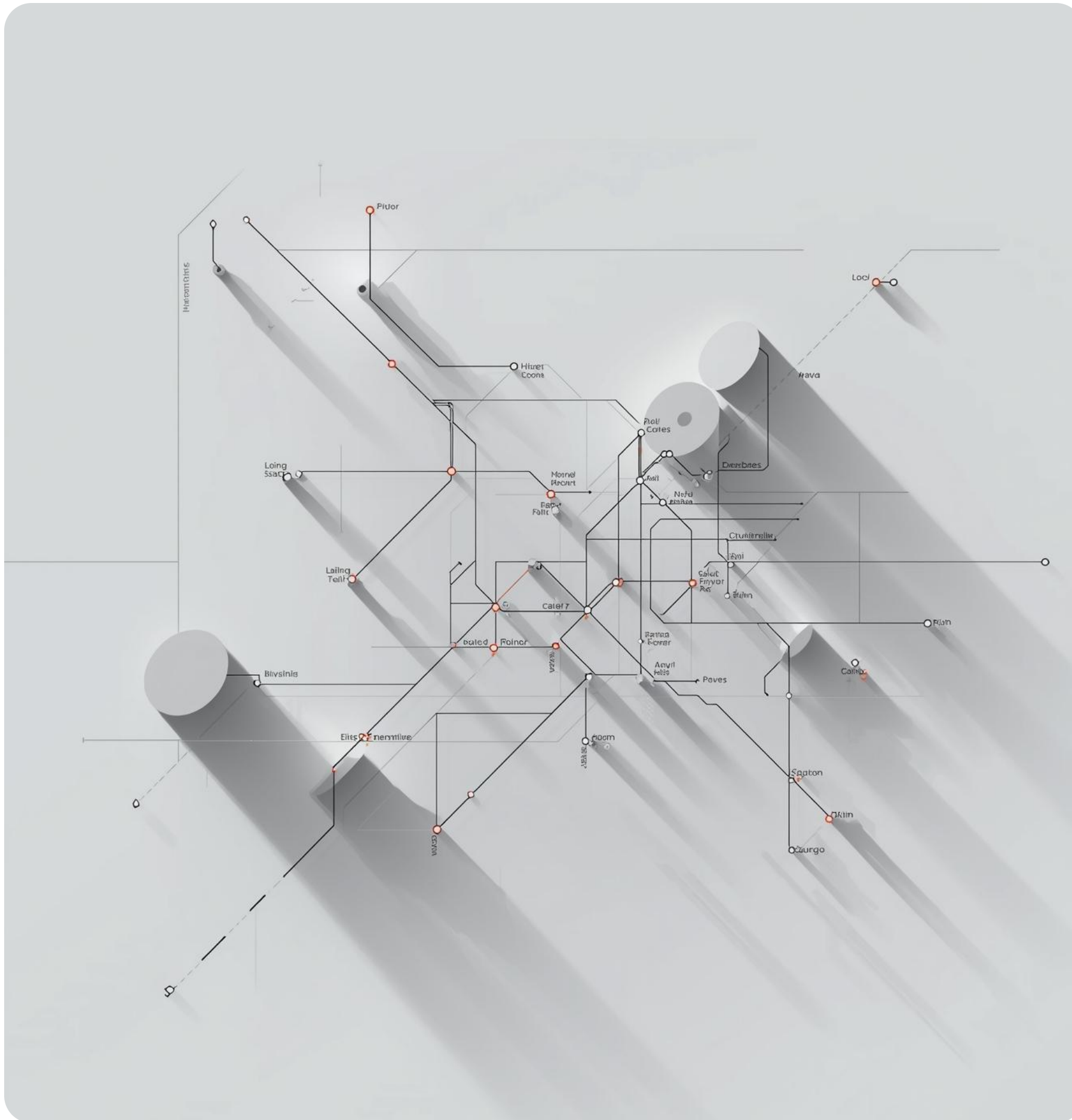


15 Oct 2025

Demographic Analysis

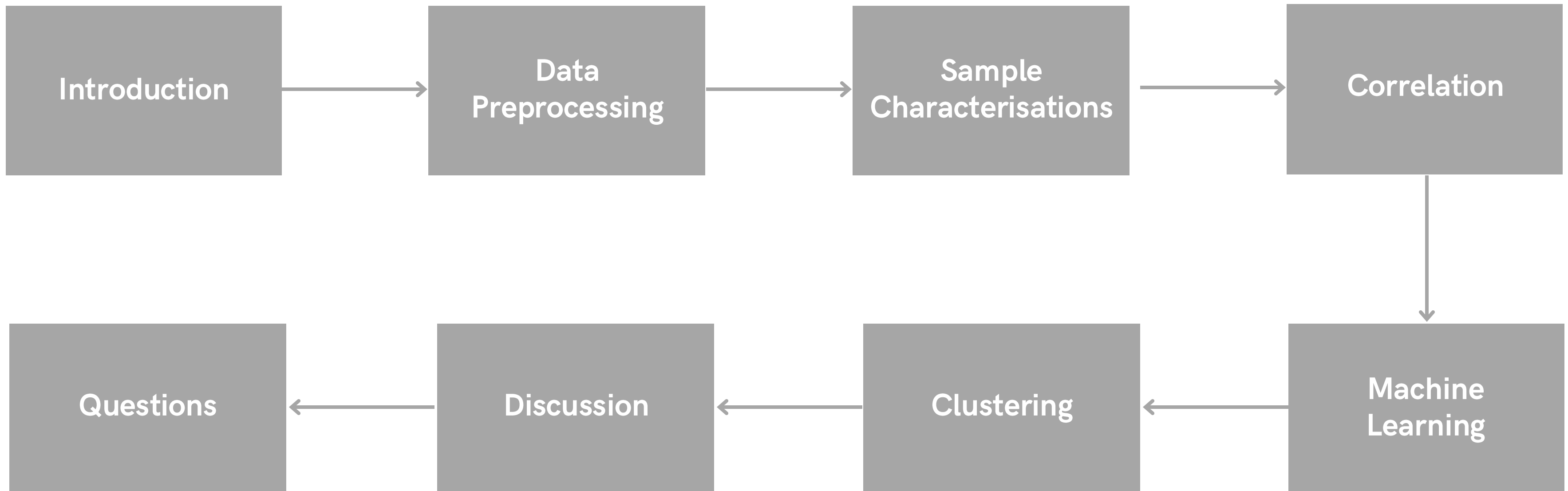
What
sociodemographic
features are
important in
predicting
travel mode?

**Peiran Wang, Shi Jian Chan, Noah Ryan,
Gerard Czerwik**



Project Overview

Investigating Demographics and Travel Mode Choices



Project Objective

Population Growth – Increasing Infrastructure demands in Victoria

1. Can the broad travel mode of the population be predicted from the present sociodemographic features?
2. Can the student population's travel mode be accurately predicted using demographic features?
3. What about the workforce population?

Key variables: Age, Occupation, Education Level, Household Income, Home Suburb, Vehicle Ownership



Dataset Preprocessing

01 Merge

Combine raw
VISTA tables

02 Mode

Identify Main
Transport Mode

03 Define

Establish Key
Outcome Variables

04 Collapse

Group categories
into bins

05 Duration

Calculate total
journey time

06 Drop

Remove unwanted
columns



Preprocessing Reasoning

01 **Mode**

Selecting the main mode reduces ambiguity and edge cases.

03 **Duration**

Maximizing coverage while being resistant to outliers is crucial.

02 **Categories**

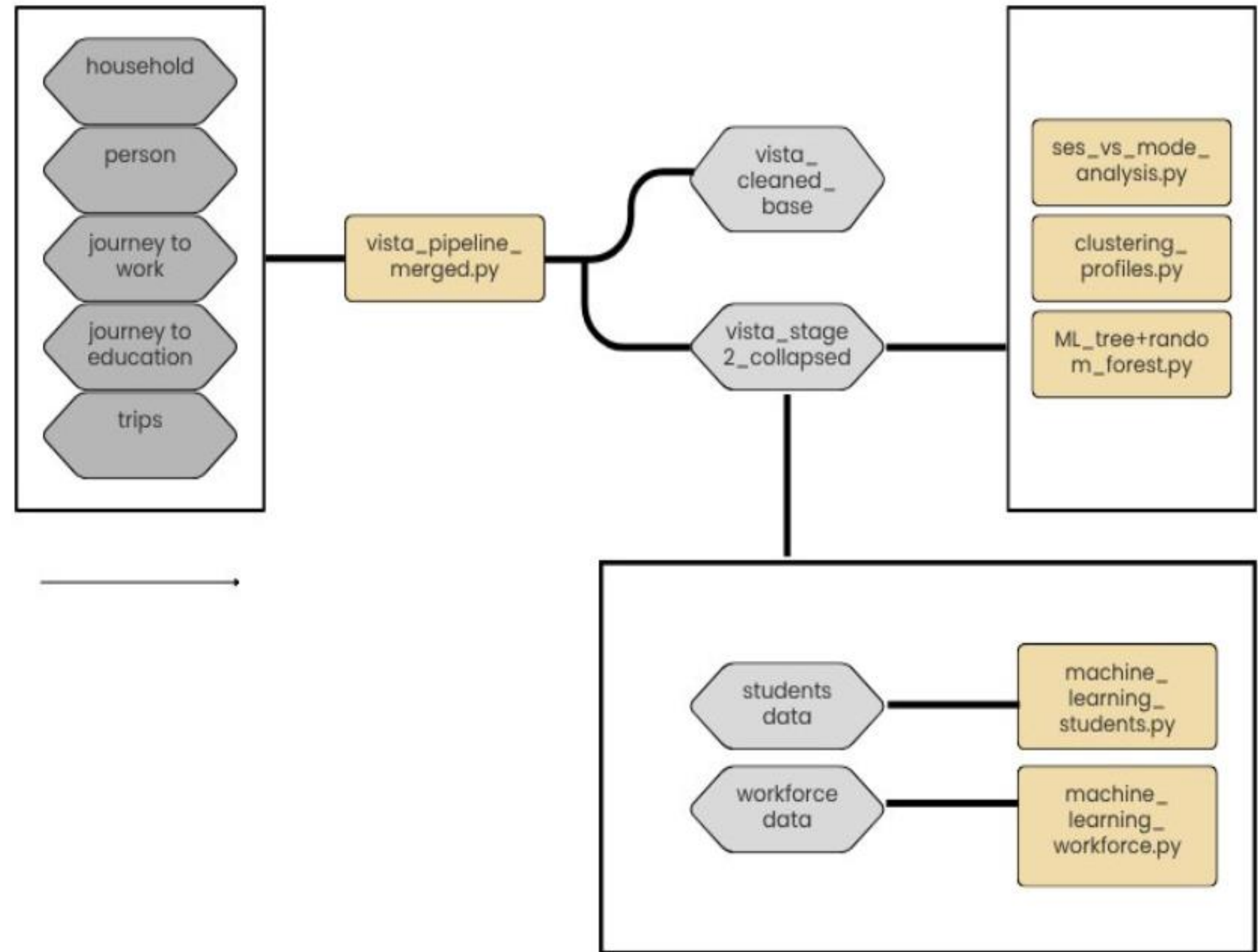
Collapsing categories leads to less sparsity and clearer statistics.

04 **Guards**

Header normalization and type-safe IDs **prevent data leakage.**

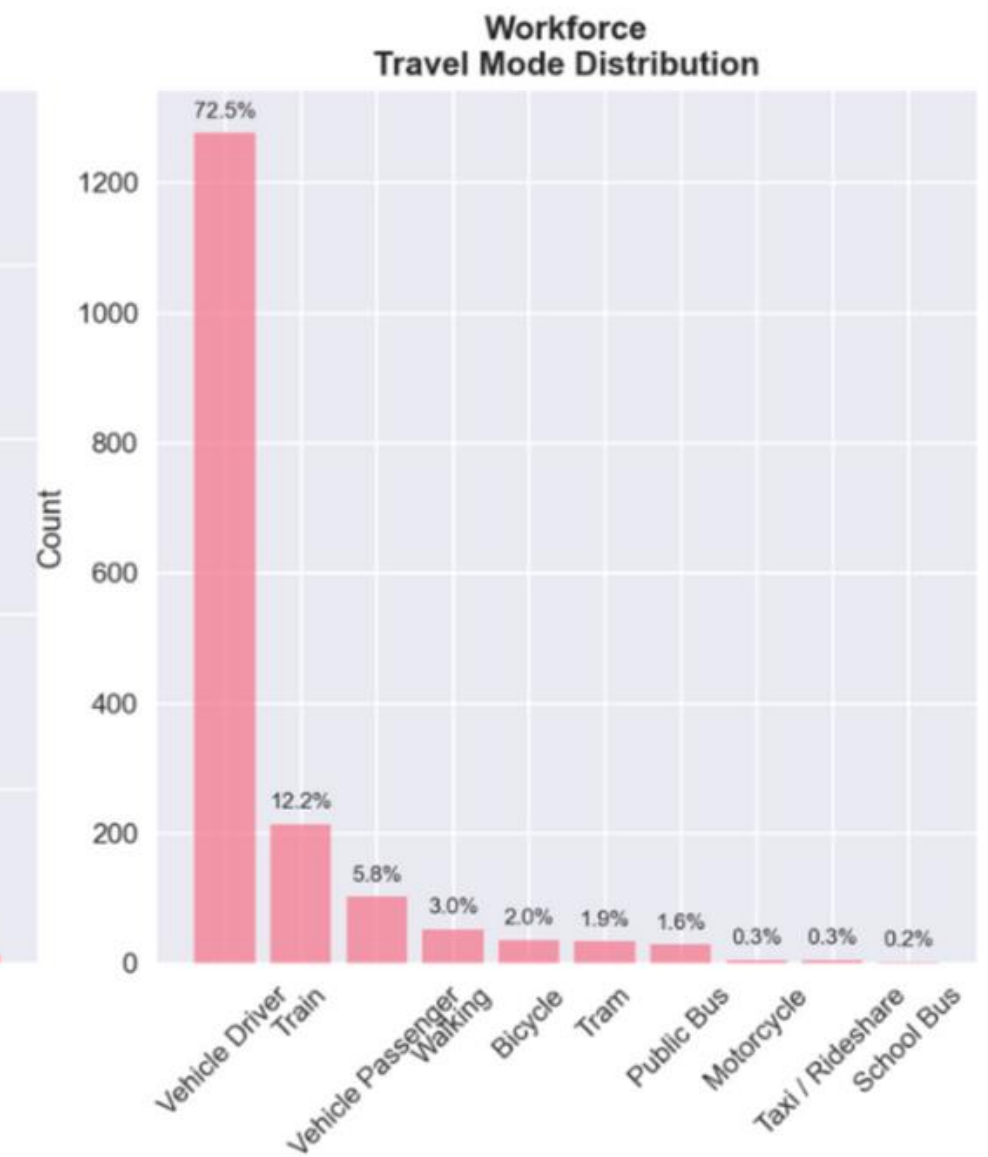
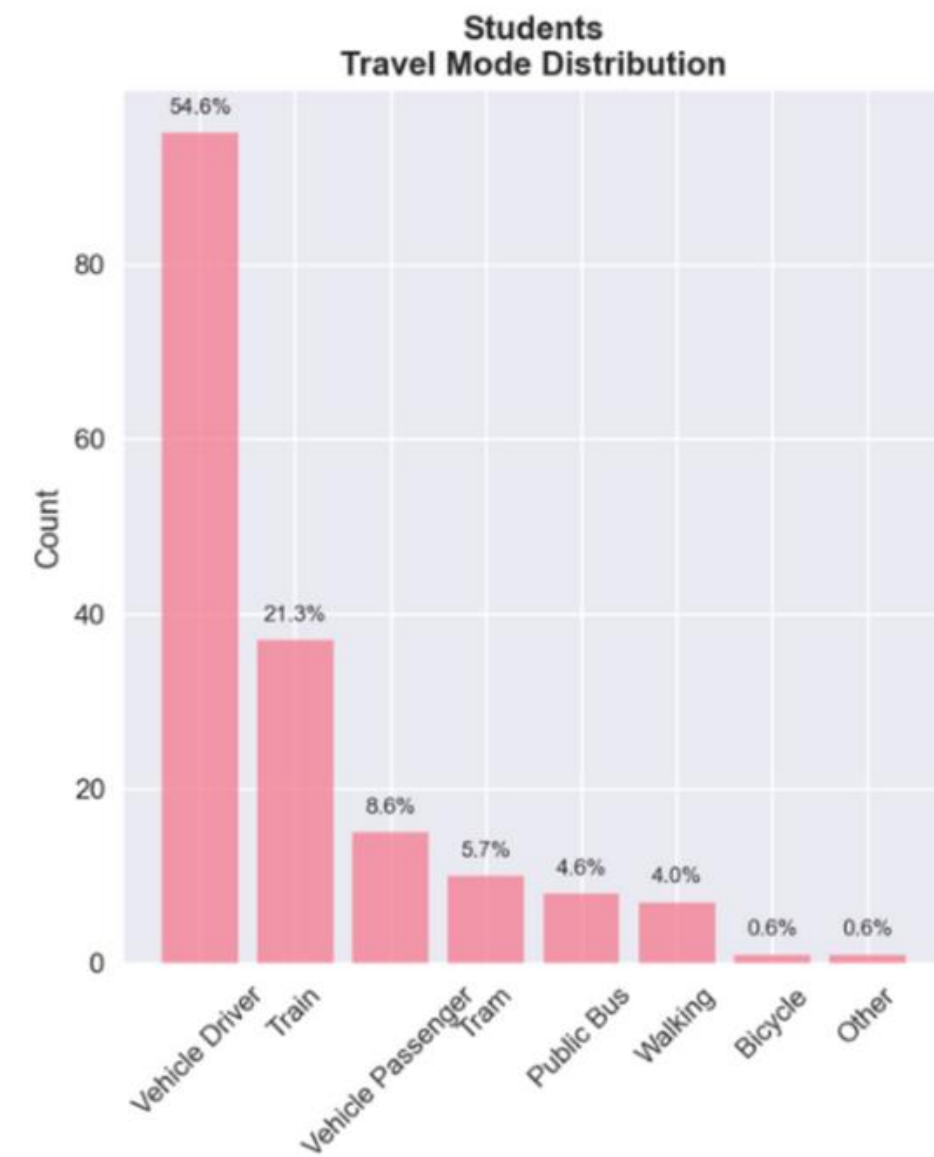
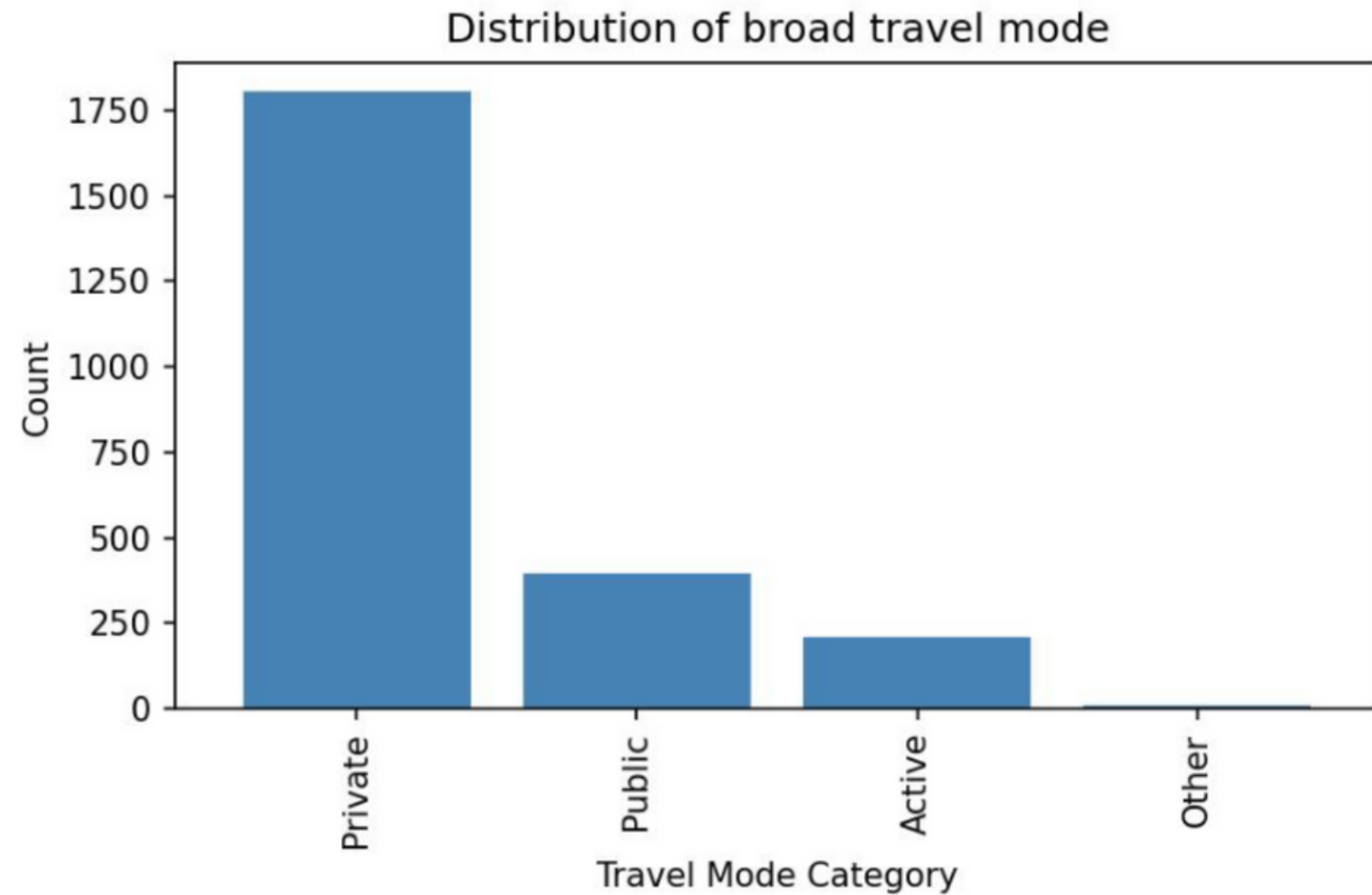
Dataflow

Data flowchart: From raw to clean



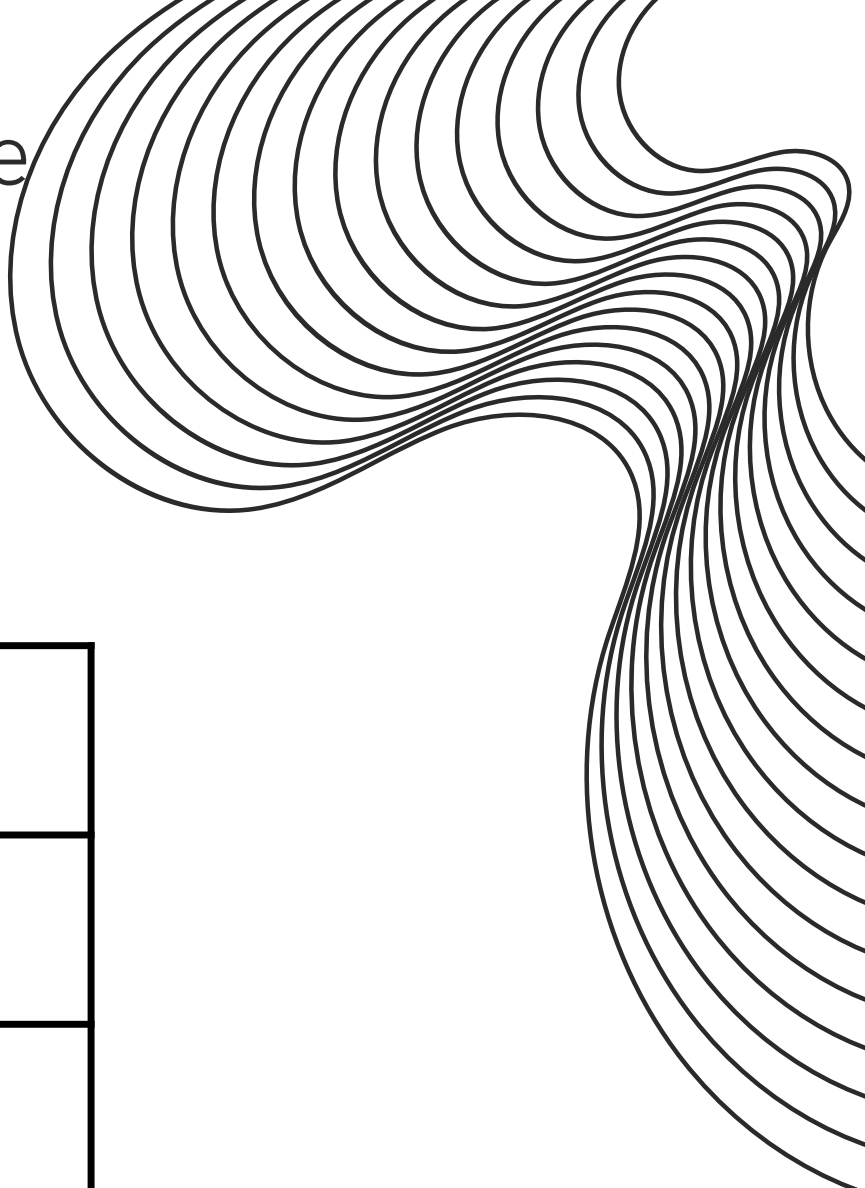
Sample Characteristics

- Workforce = 1759, Mode Age 35-39 and 50-54
- Student = 174, Mode Age 20-24



Correlation

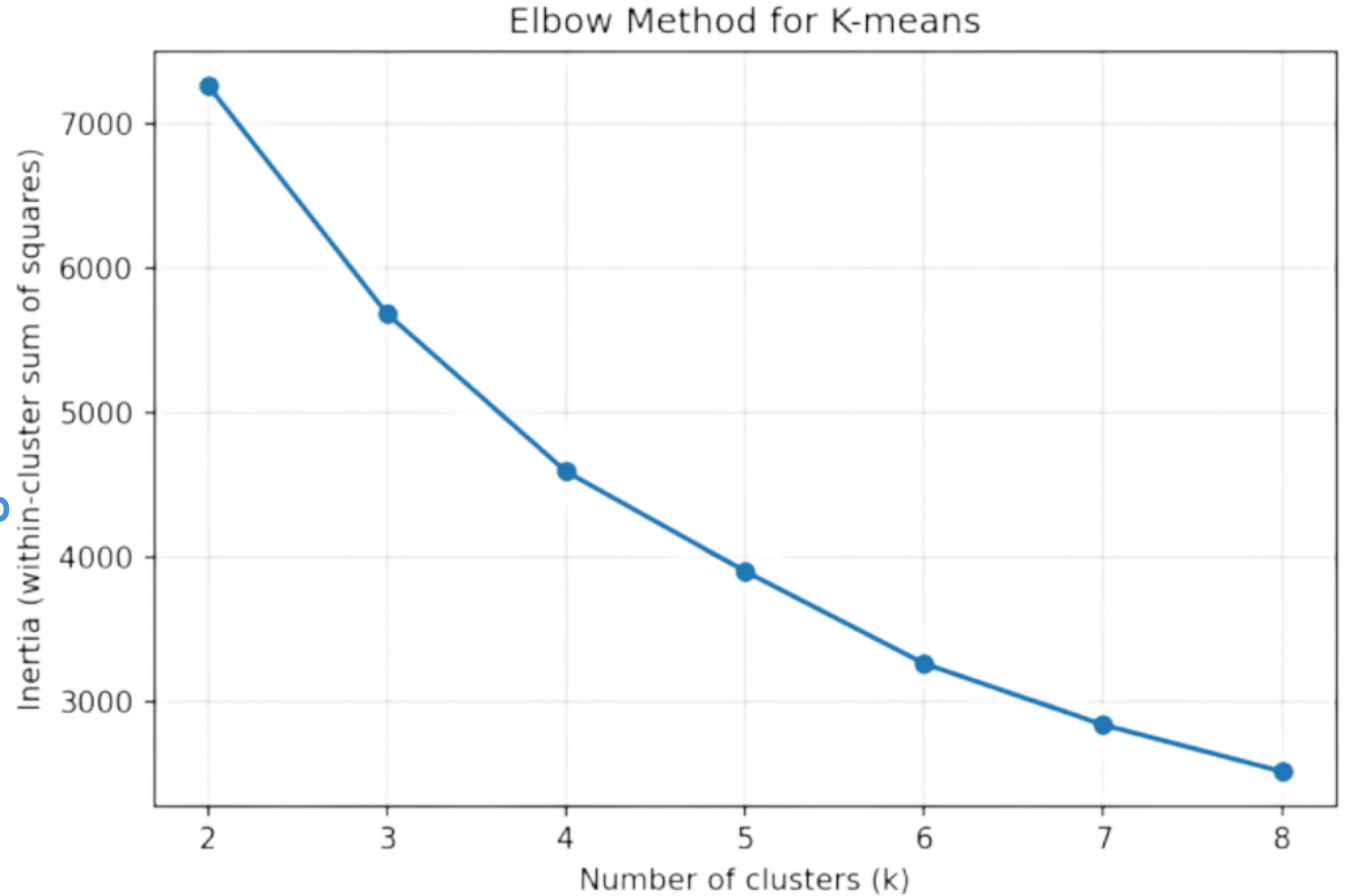
- Broad Travel Mode vs Detailed Travel Mode
- Weaker correlations for Broad
- Medium Associations



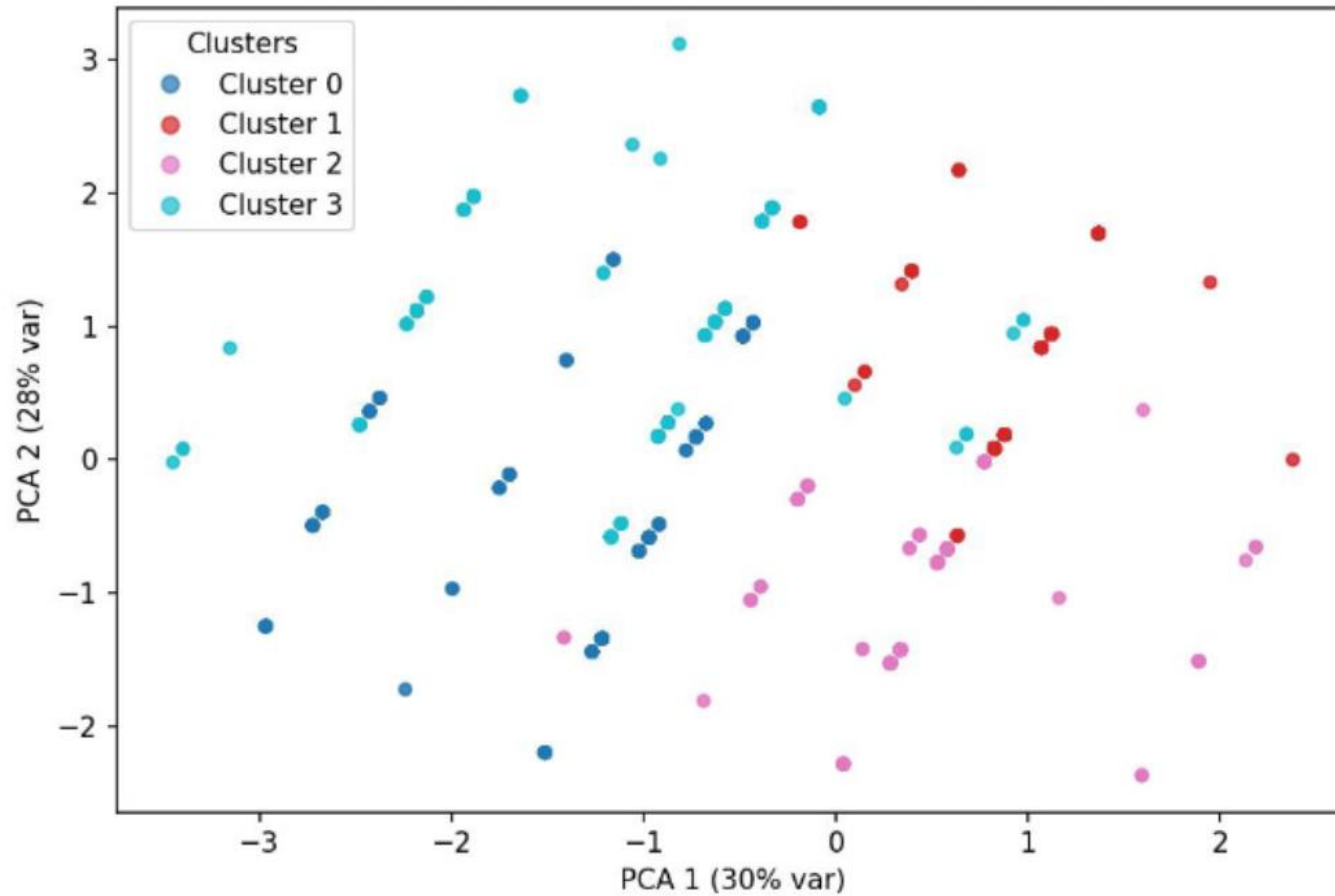
Predictor	Cramér's V	NMI
Vehicle Ownership	0.29	0.21
Home Location	0.26	0.20
Household Income	0.24	0.18
Education Status	0.17	0.12
Occupation	0.15	0.11
Age Stage	0.13	0.10

Clustering

Inputs: occupation, income,
region, vehicles owned, suburb



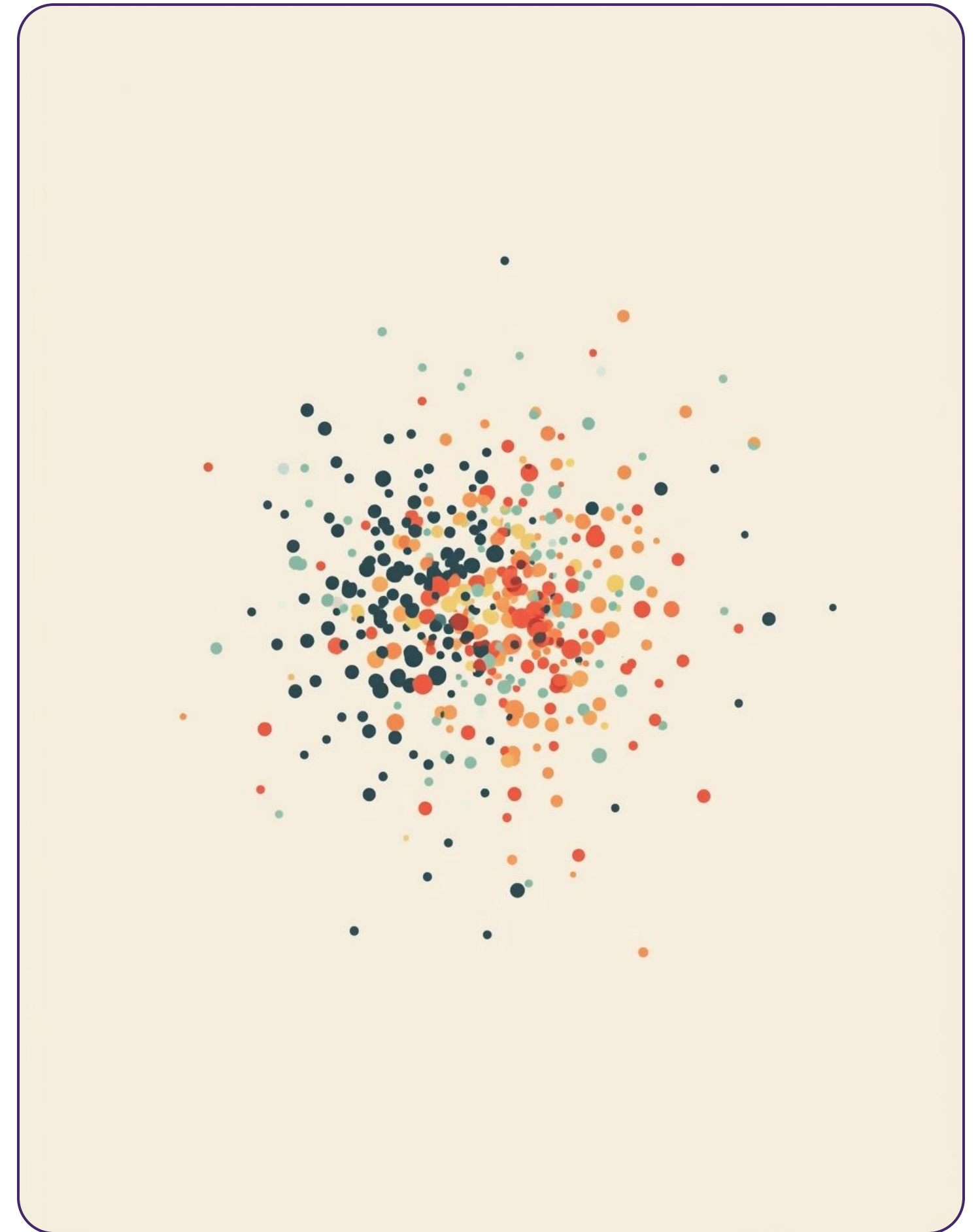
K-means clusters of workforce travel profiles (k=4)



Machine Learning

Employed 3 models with variations for the completion of 3 different studies:

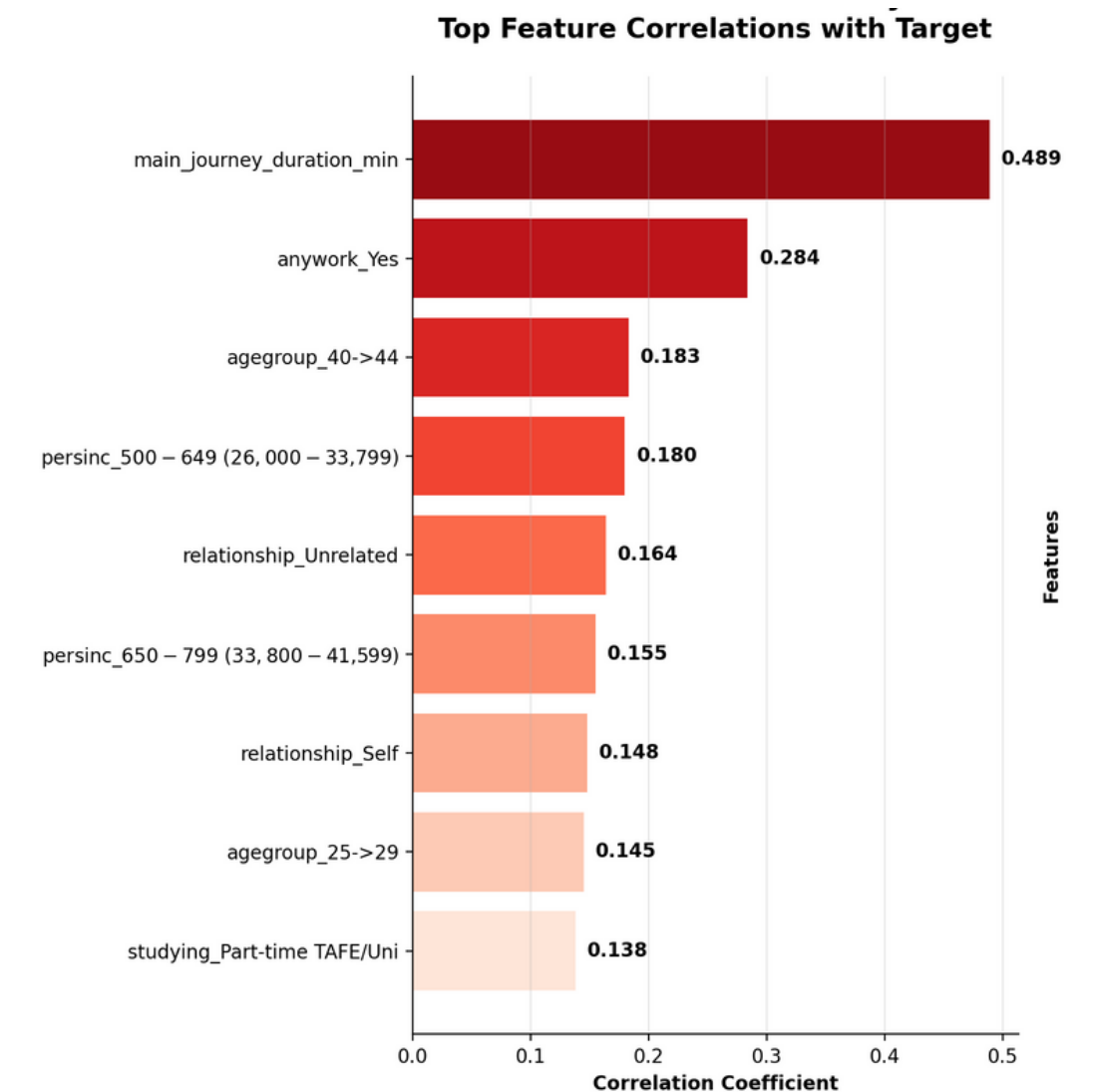
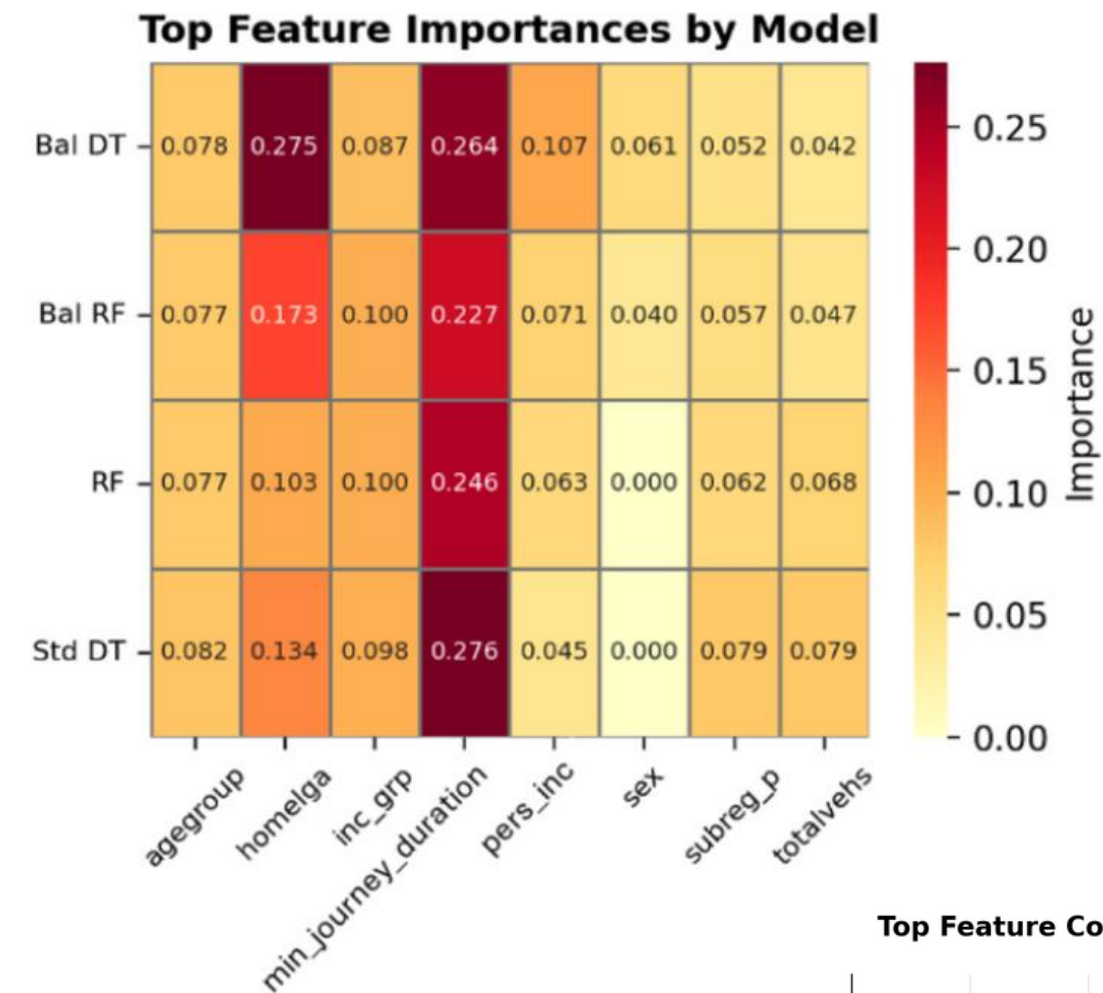
1. Can the broad travel mode of the population be predicted from the present demographic features?
2. Can the student population's travel mode be accurately predicted using demographic features?
3. What about the workforce population?



Choosing and Preparing Models

Decision Tree, Random Forest and Logistical Regression

- Pairwise comparisons with Gini index or otherwise correlation analysis for feature selection
- K fold cross validation for hyperparameter tuning of most optimal models from feature selection

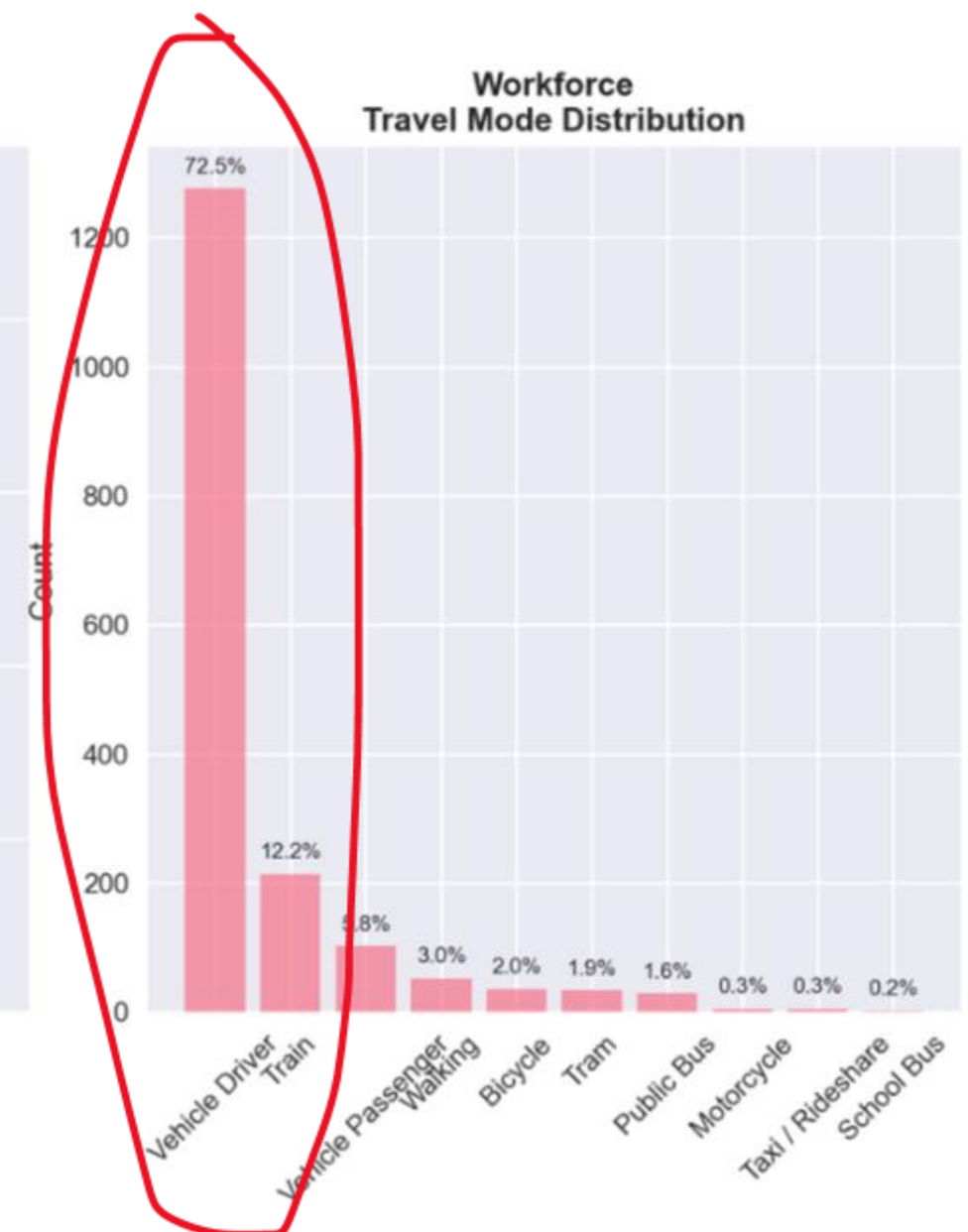
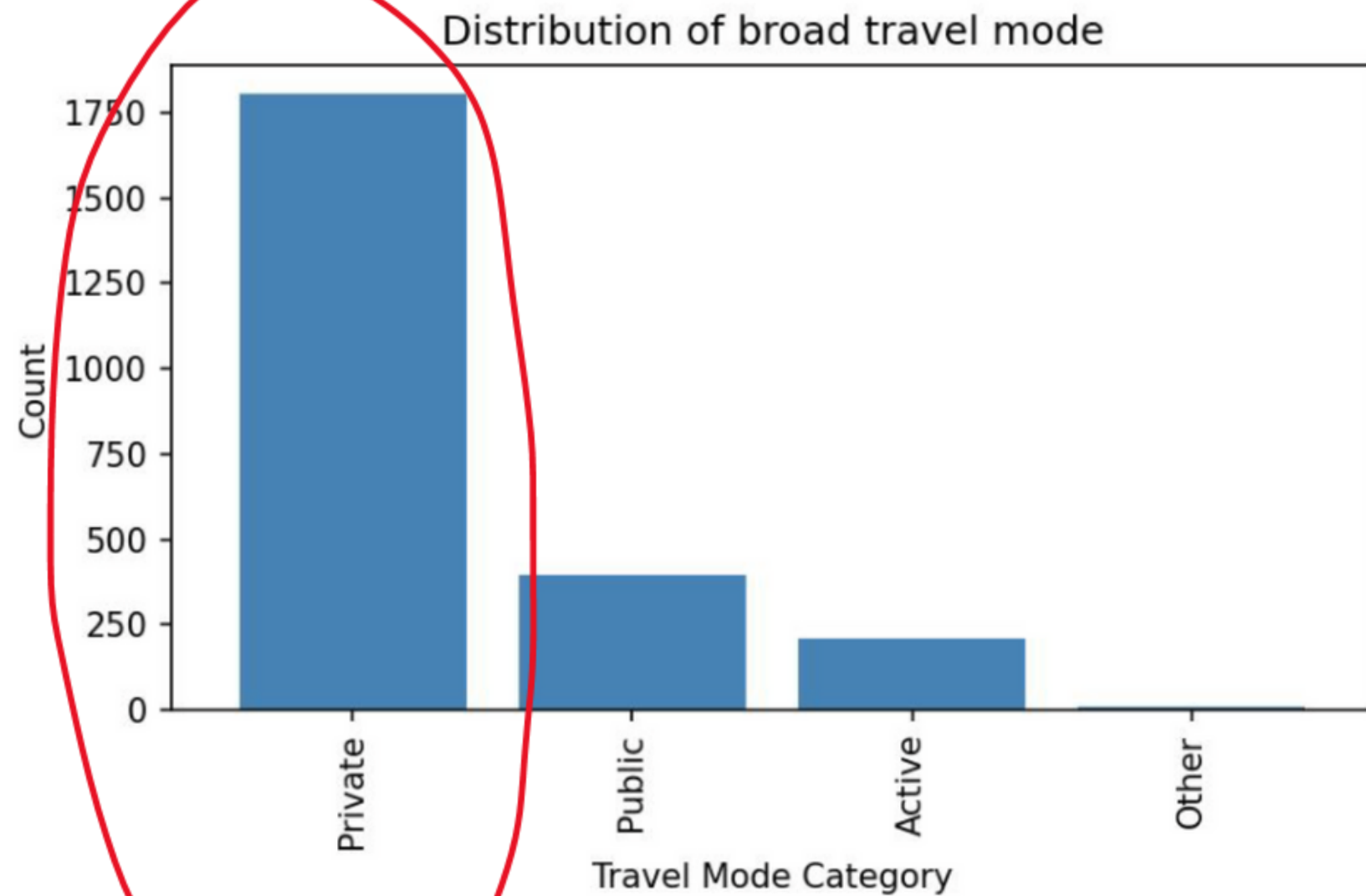


Results

Question Type	Model Type	Accuracy	Balanced Accuracy	F1 Macro
Overall Population	Random Forest	0.796	0.360	0.415
Students	Random Forest	0.800	0.500	0.458
Workforce	Logistical Regression	0.770	0.302	0.321

Sample Characteristics

- Workforce = 1759, Mode Age 35-39 and 50-54
- Student = 174, Mode Age 20-24



Discussion

Evaluating strengths, limitations, and applications

Patterns can help create targeted interventions address needs targeting different sociodemographic clusters.



Questions

